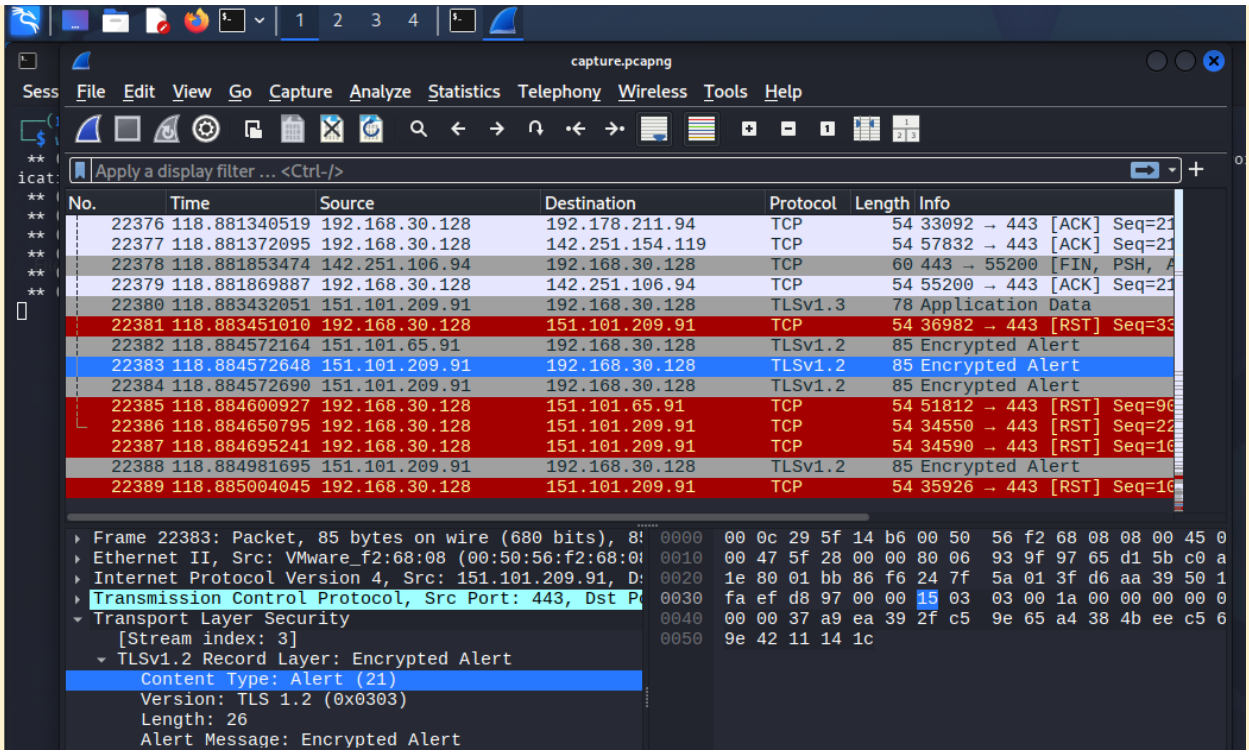
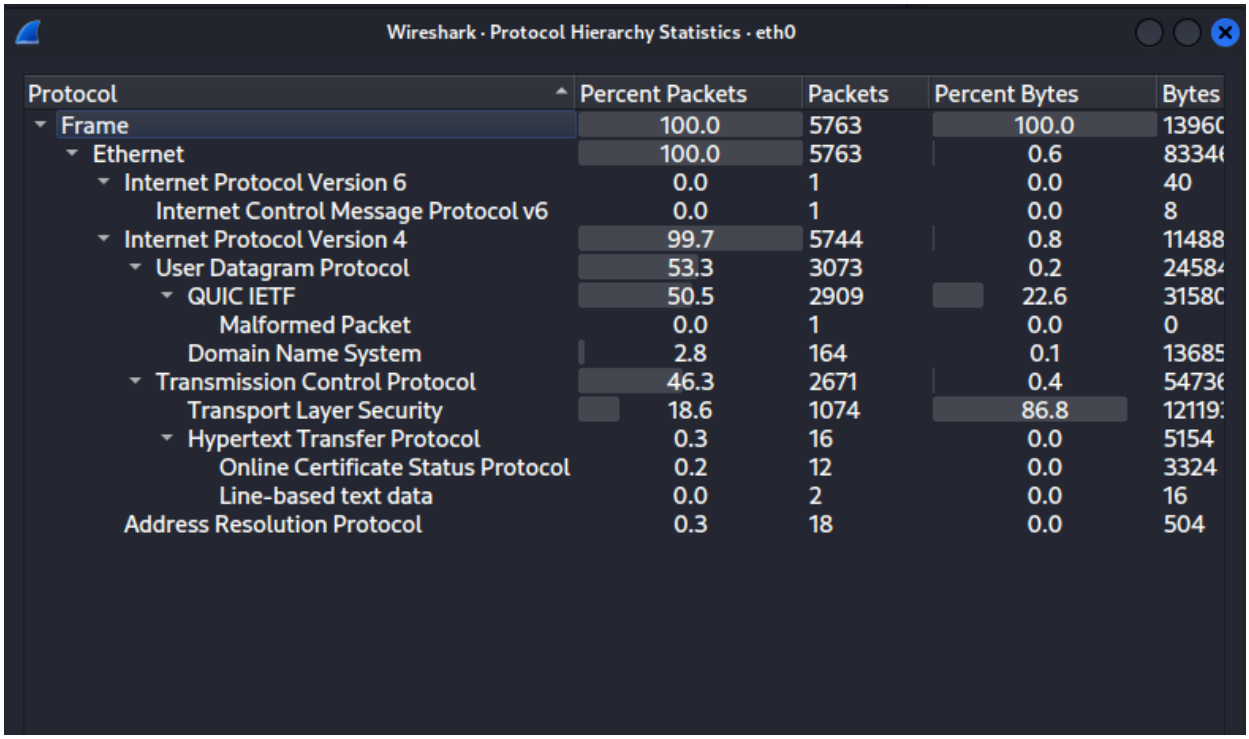
Software and Hardware Tools

Task 1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Answer:

1. Download Wireshark from the official website (wireshark.org) and install it (also allow the Npcap driver when it asks).
2. Open Wireshark and double-click on your active network (Wi-Fi or Ethernet). The capture starts, and you will see packets moving.
3. Open Instagram or YouTube in your browser and browse for 2 minutes.
4. Click the red Stop button.
5. Save the capture: File > Save As, choose the file type "Wireshark/tcpdump/... - pcap", and name it, for example capture.pcap.
6. Find the top protocols: Statistics > Protocol Hierarchy. Check the 3 protocols with the highest percentage of packets.





My capture details (from Statistics > Capture File Properties):

Item	Value
File name	capture.pcapng (to be saved also as .pcap, see note below)
Capture tool	Wireshark 4.6.6 (Dumpcap), interface eth0, Ethernet
First packet	2026-11-06 08:48:14
Last packet	2026-11-06 08:50:13
Duration (elapsed)	00:01:58 (about 2 minutes)
Total packets captured	22,389 (no dropped packets)
File size / total bytes	31 MB (31,242,205 bytes)
Average packet size	1395 bytes
Average speed	188.3 packets per second, about 2.1 Mbps

My top 3 protocols (write from your own capture):

No.	Protocol seen in my capture	What it is used for
1	QUIC (IETF), 74.0% of packets (16,573 packets)	Fast, encrypted web transport that runs over UDP. YouTube and Instagram use it.
2	TCP, 24.8% of packets (5,550 packets)	Reliable data transfer between my computer and the servers; carries the TLS / HTTPS traffic.
3	TLS, 9.8% of packets (2,195 packets)	Encrypts the data of secure websites (HTTPS) on top of TCP.

Result (Statistics > Protocol Hierarchy): QUIC is the top protocol with 74.0% of the packets. UDP (75.1%) is the transport layer that carries QUIC, and DNS (1.1%, 242 packets) was also seen for name lookups. TCP is the next highest with 24.8%, and TLS runs on it with 9.8%.

Plain HTTP was almost not seen (only 2 packets), because most sites now use encrypted HTTPS / QUIC.

Task 2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Answer:

```
ping www.spotify.com
ping www.zomato.com
```

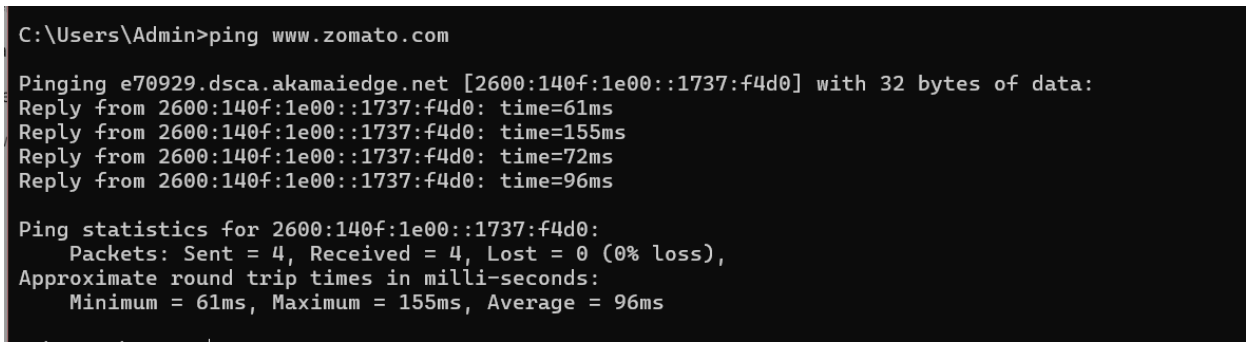
At the end of each ping, write the Minimum, Maximum and Average times. The site with the lower Average time responds faster.

Website	Minimum (ms)	Maximum (ms)	Average (ms)	Packet loss
www.spotify.com	55	187	94	0% (4 sent, 4 received)
www.zomato.com	61	155	96	0% (4 sent, 4 received)

Spotify result: the ping resolved www.spotify.com to atc.spotify.map.fastly.net (a Fastly CDN server) and replied over IPv6 (2a04:4e42:31::810). All 4 packets got a reply (0% loss), with times of 61, 187, 55 and 74 ms. Zomato result: www.zomato.com resolved to e70929.dsca.akamaiedge.net (an Akamai CDN server) and also replied over IPv6 (2600:140f:1e00::1737:f4d0). All 4 packets got a reply (0% loss), with times of 61, 155, 72 and 96 ms.

Faster site: www.spotify.com responds faster, with an average of 94 ms compared to 96 ms for www.zomato.com. The difference is only 2 ms, so both sites respond at almost the same speed. Both use CDN servers (Fastly for Spotify and Akamai for Zomato), which are close to my network.





Task 3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Answer:

1. Press Ctrl + Shift + Esc to open Task Manager (on Mac, open Activity Monitor from Spotlight).
2. Open Chrome (and WhatsApp Desktop if you use it) so that they are running.
3. In the Processes tab, find "Google Chrome" and expand it with the small arrow to see its sub-processes. You can also use the Details tab to see each chrome.exe.
4. Write the process name and the Memory column value for each. Use the Memory column to sort from the highest.

No.	Process name	Memory usage (MB)
1	Google Chrome (Efficiency mode)	518.0 MB
2	Google Chrome (Efficiency mode)	172.4 MB
3	Google Chrome	107.5 MB
4	Google Chrome	88.1 MB
5	Google Chrome (Efficiency mode)	70.0 MB
6	Google Chrome (Efficiency mode)	54.0 MB
7	Google Chrome (Efficiency mode)	21.9 MB
8	Google Chrome	16.5 MB
9	Google Chrome (Efficiency mode)	9.0 MB
10	Google Chrome	7.1 MB
11	Google Chrome	2.0 MB
12	Google Chrome	1.8 MB
13	Google Chrome	0.6 MB
	Total of the 13 Chrome processes	1,068.9 MB (about 1.04 GB)

Processes

Name	Status	10% CPU	96% Memory	1% Disk	0% Network
Google Chrome	Efficiency ...	0%	518.0 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency ...	0%	172.4 MB	0 MB/s	0 Mbps
Google Chrome		0%	107.5 MB	0 MB/s	0 Mbps
Google Chrome		0%	88.1 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency ...	0%	70.0 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency ...	0%	54.0 MB	0 MB/s	0 Mbps
Google Chrome	Efficiency ...	0%	21.9 MB	0 MB/s	0 Mbps
Google Chrome		0%	16.5 MB	0.1 MB/s	0 Mbps
Google Chrome	Efficiency ...	0%	9.0 MB	0 MB/s	0 Mbps
Google Chrome		0%	7.1 MB	0 MB/s	0 Mbps
Google Chrome		0%	2.0 MB	0 MB/s	0 Mbps
Google Chrome		0%	1.8 MB	0 MB/s	0 Mbps
Google Chrome		0%	0.6 MB	0 MB/s	0 Mbps
ChatGPT (13)		0%	529.6 MB	0 MB/s	0.1 Mbps
Codex		0%	245.9 MB	0 MB/s	0 Mbps
Codex	Efficiency ...	0%	148.8 MB	0 MB/s	0 Mbps
Codex		0%	92.5 MB	0 MB/s	0 Mbps
Codex		0%	9.9 MB	0 MB/s	0 Mbps
Codex		0%	9.2 MB	0 MB/s	0 Mbps
Codex		0%	8.2 MB	0 MB/s	0.1 Mbps
Codex		0%	4.7 MB	0 MB/s	0 Mbps
Codex	Efficiency ...	0%	3.3 MB	0 MB/s	0 Mbps
Codex	Efficiency ...	0%	2.4 MB	0 MB/s	0 Mbps
Codex		0%	1.9 MB	0 MB/s	0 Mbps

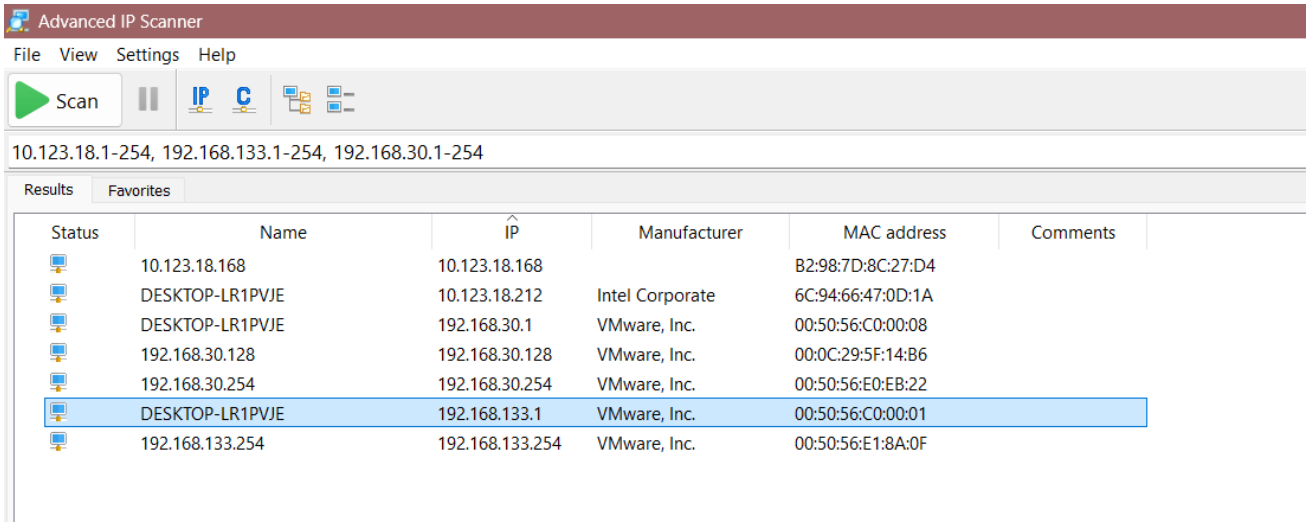
Result: Task Manager shows 13 Google Chrome processes (browser, tabs, extensions and background services). The biggest one uses 518.0 MB and the smallest uses 0.6 MB. Together they use about 1,068.9 MB. The processes marked with the green leaf use Efficiency mode, which reduces the resources of background processes. The screenshot also shows that the total system memory use was 96%.

Task 4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Answer:

1. Download Angry IP Scanner (angryip.org) or Advanced IP Scanner from the official website and install or run it.
2. Find your network range first: run ipconfig and look at your IPv4 address (for example 192.168.1.15), so the range is 192.168.1.1 to 192.168.1.254.
3. In the scanner, enter this IP range (many scanners fill it automatically) and click Start / Scan.
4. Wait until the scan finishes. The live devices appear with their IP address and hostname (device name).
5. Write at least 2 devices from the list.

No.	IP address	Device name / hostname	Device type (phone, laptop, router...)
1	10.123.18.212	DESKTOP-LR1PVJE	My Windows laptop on the Wi-Fi network (manufacturer: Intel Corporate, MAC 6C:94:66:47:0D:1A)
2	10.123.18.168	No name shown (IP only)	Another device on the same Wi-Fi network, probably the router / hotspot (MAC B2:98:7D:8C:27:D4)
3	192.168.30.128	192.168.30.128	Kali Linux virtual machine in VMware (MAC 00:0C:29:5F:14:B6)
4	192.168.30.1	DESKTOP-LR1PVJE	My laptop's VMware virtual adapter (MAC 00:50:56:C0:00:08)
5	192.168.30.254	192.168.30.254	VMware virtual network gateway / DHCP (MAC 00:50:56:E0:EB:22)
6	192.168.133.1	DESKTOP-LR1PVJE	My laptop's second VMware virtual adapter (MAC 00:50:56:C0:00:01)
7	192.168.133.254	192.168.133.254	VMware virtual network gateway (MAC 00:50:56:E1:8A:0F)



Result: Advanced IP Scanner scanned three ranges (10.123.18.1-254, 192.168.133.1-254 and 192.168.30.1-254) and found 7 live devices. The first two devices (10.123.18.212 and 10.123.18.168) are on my real Wi-Fi network. The other five addresses (192.168.30.x and 192.168.133.x) belong to the virtual networks created by VMware on my laptop, which is why their manufacturer is VMware, Inc.

Task 5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Answer:

No.	Hardware tool	When I would use it (one line)
1	Network cable tester	When a computer shows "network cable unplugged" or has no connection, I use it to check whether an Ethernet cable has a broken wire, a wrong wire order or a loose connector.
2	Crimping tool (with RJ45 connectors)	When I need to make a new LAN cable or replace a damaged RJ45 connector on the end of a cable.
3	Multimeter	When I need to check power problems, for example to test the voltage of a power adapter or a UPS, or to check continuity of a cable.